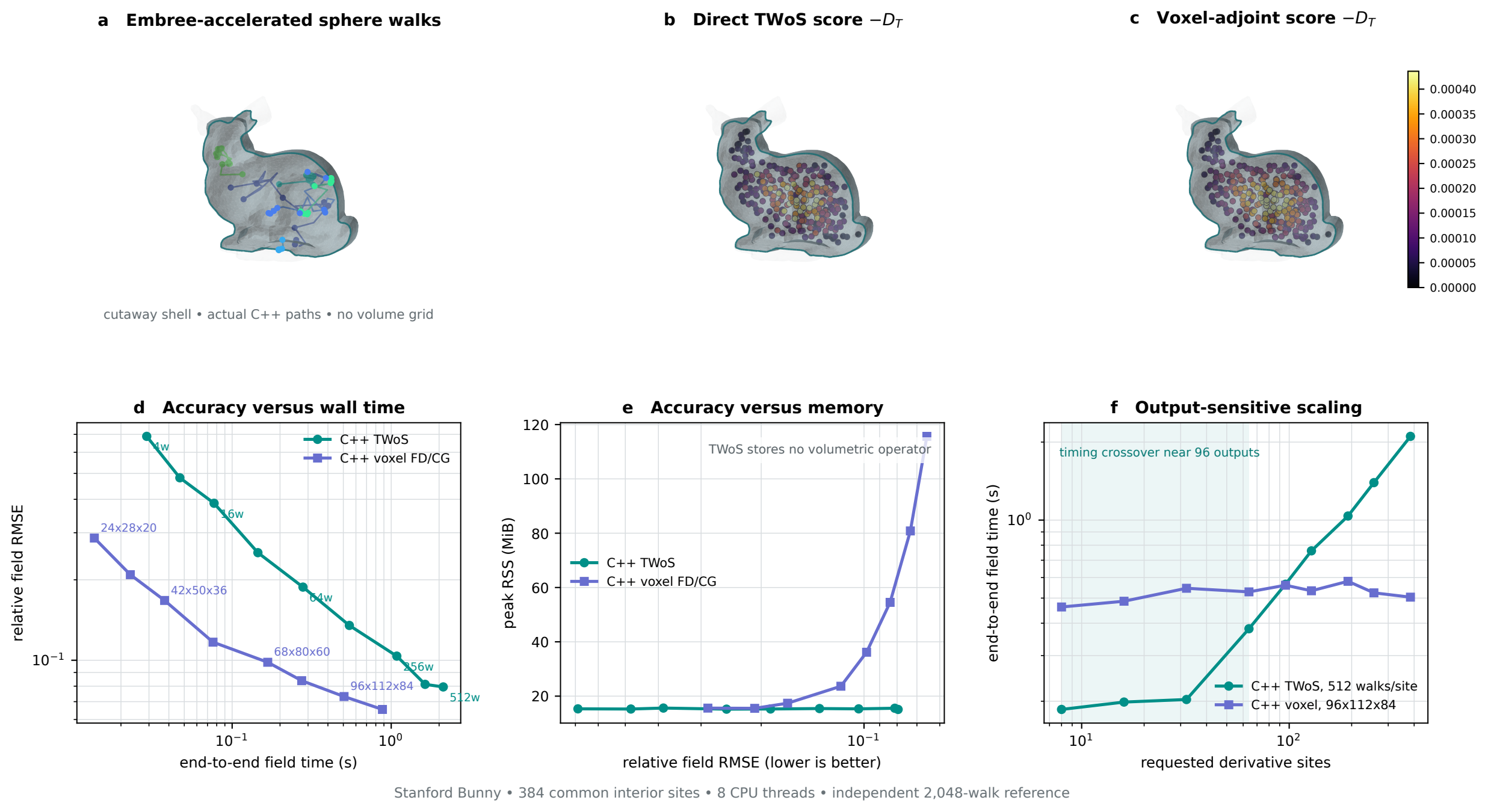

a Embree-accelerated sphere walks
b Direct TWoS score $-D_T$
c Voxel-adjoint score $-D_T$
0.00040
0.00035
0.00030
0.00025
0.00020
0.00015
0.00010
0.00005
0.00000
cutaway shell • actual C++ paths • no volume grid
d Accuracy versus wall time
C++ TWoS
C++ voxel FD/CG
1w
16w
64w
256w
512w
24x28x20
42x50x36
68x80x60
96x112x84
relative field RMSE
end-to-end field time (s)
e Accuracy versus memory
TWoS stores no volumetric operator
C++ TWoS
C++ voxel FD/CG
peak RSS (MiB)
relative field RMSE (lower is better)
f Output-sensitive scaling
timing crossover near 96 outputs
C++ TWoS, 512 walks/site
C++ voxel, 96x112x84
end-to-end field time (s)
requested derivative sites
Stanford Bunny • 384 common interior sites • 8 CPU threads • independent 2,048-walk reference